

1.1 Overview of JDemetra+

ESTP Training
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eurostat 

0. Outline

- Objectives of JDemetra+ (JD+)
 - General
 - For seasonal adjustment (SA)
- What is really JD+ ?
- Statistical content
- Seasonal adjustment framework
- Final remarks

1.1 General Objectives

- Providing algorithms for the production/analysis of [official] statistics
 - Regular time series (from monthly to yearly)
 - Algorithms for
 - Seasonal adjustment, business cycle analysis
 - Benchmarking, temporal disaggregation
 - Modelling (forecasting, estimation of missing values, outliers detection)
- Reusable modules, compatible with common IT infrastructure
 - Java, R, WEB services...
- Designed for the whole statistical process
 - From research to bulk production (flexible, high-performance)
- Maintainable
 - Open-source solution

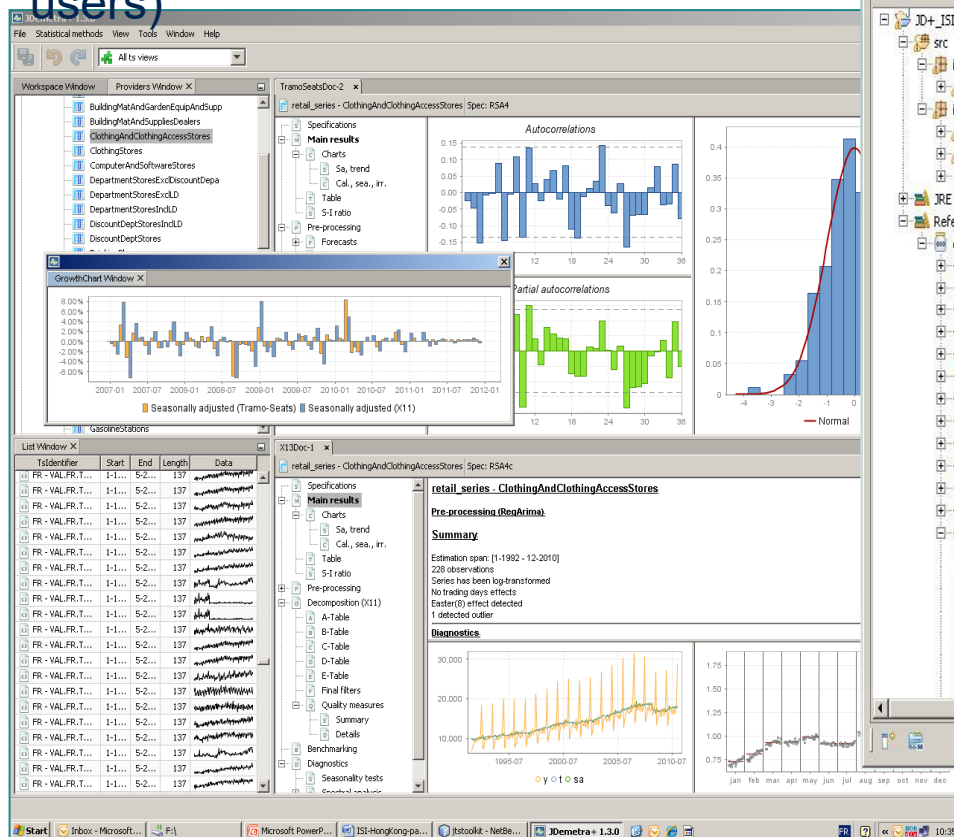
1.2 Objectives for SA

- Java implementation of the leading algorithms
 - TRAMO-SEATS, X12-ARIMA...
- Flexible design
 - Easier modifications of the core engines
 - Developments of additional tools/algorithms
- Challenge
 - Keeping similar results and high performances
 - with flexible (more general) design and algorithms and slower technical solution

2.1 What is JD+ (I) ?

Advanced Java toolkit for time series
(SA) processing (IT-teams,
researchers)

Rich graphical application (end-users)



The screenshot shows the Java source code editor for the JD+ application. The code is organized into a package structure under 'JD+_ISI'. The main code file is 'App.java', which contains the following methods:

- public static TsData series(int n):** A method that generates a time series of size n. It uses 'DataBlock' and 'TsPeriod' to create a series of random data points.
- public static UcarimaModel canonical():** A method that creates a canonical Ucarima model. It uses 'SarimaModelBuilder' to create a model, 'TrendCycleSelector' to select a trend cycle, and 'SeasonalSelector' to select a seasonal component. The model is then decomposed using 'ModelDecomposer'.
- public static TsData saMcElroy(UcarimaModel m, TsData ts):** A method that applies the McElroy algorithm to a time series. It uses 'McElroyEstimates' to calculate the seasonal component and 'TsData' to store the results.

2.2 What is JD+ ? (II)

JDemetra+
Seasonal adjustment and time series analysis tool

v3 projects dashboard

Repository	Latest release	Unreleased changes	Reproducible builds	Builds	Pull requests
jdplus-main	v3.9.0	#0	resource not found	passing	2 open
jdplus-benchmarking	v2.2.0	#0	resource not found	passing	0 open
jdplus-incubator	v2.5.2	#1	resource not found	passing	0 open
jdplus-experimental	v1.1.0	#0	resource not found	passing	0 open
jdplus-revisions	v1.1.0	#2	resource not found	passing	0 open
jdplus-nowcasting	v2.0.0	#2	resource not found	passing	0 open
jdplus-examples	-	#0	resource not found	passing	2 open
jdplus-ws	-	#0	resource not found	passing	1 open
jdplus-eurostat	-	#0	resource not found	passing	1 open

People

Top languages
Java JavaScript HTML

Most used topics
jdemetra-plugin seasonal-adjustment command-line-tool desktop-application jdemetra

Open Source project (EUPL license)

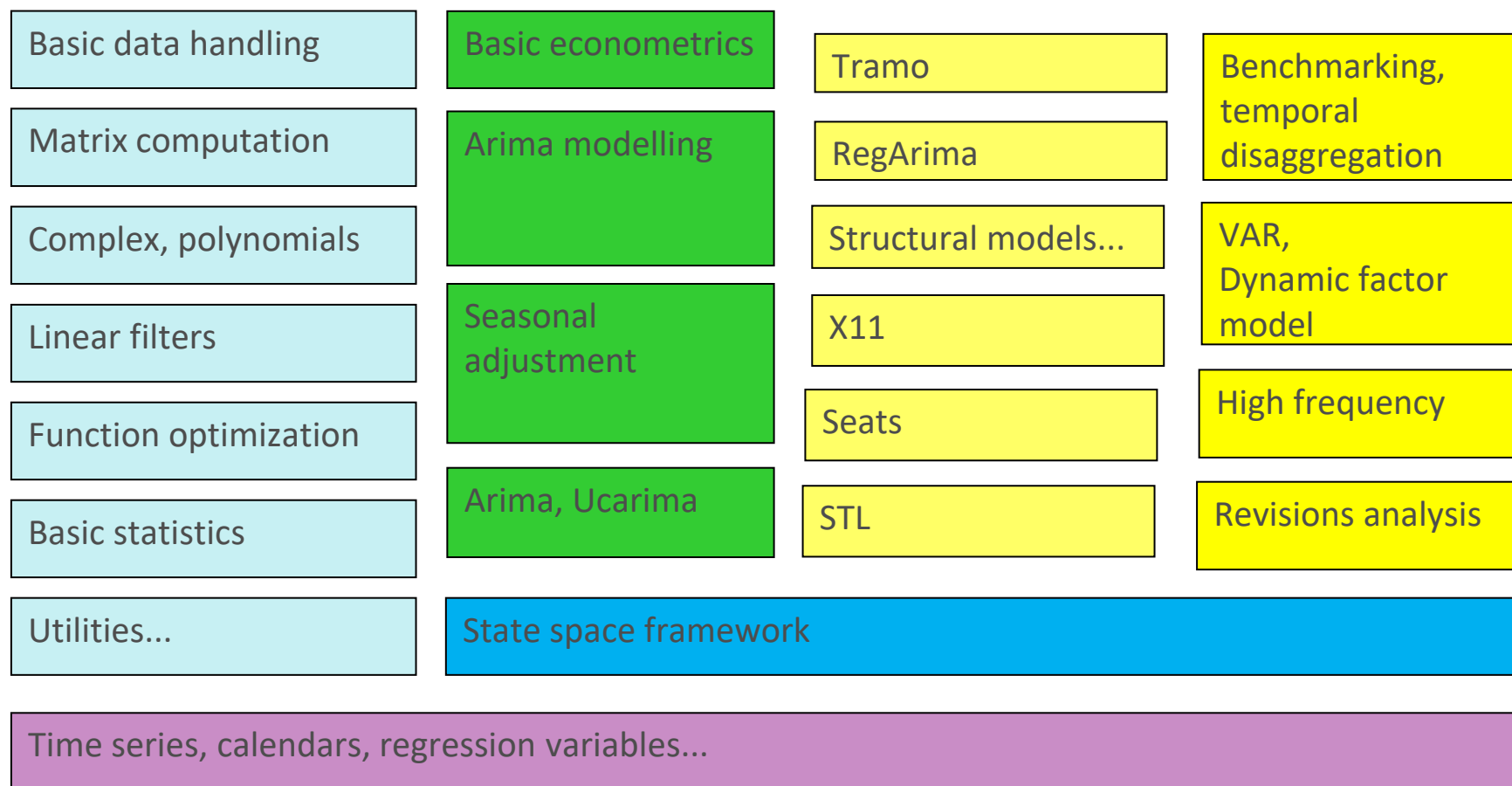
- Supported by Eurostat

- Developers:
- NBB
- Bundesbank
- INSEE
- ...

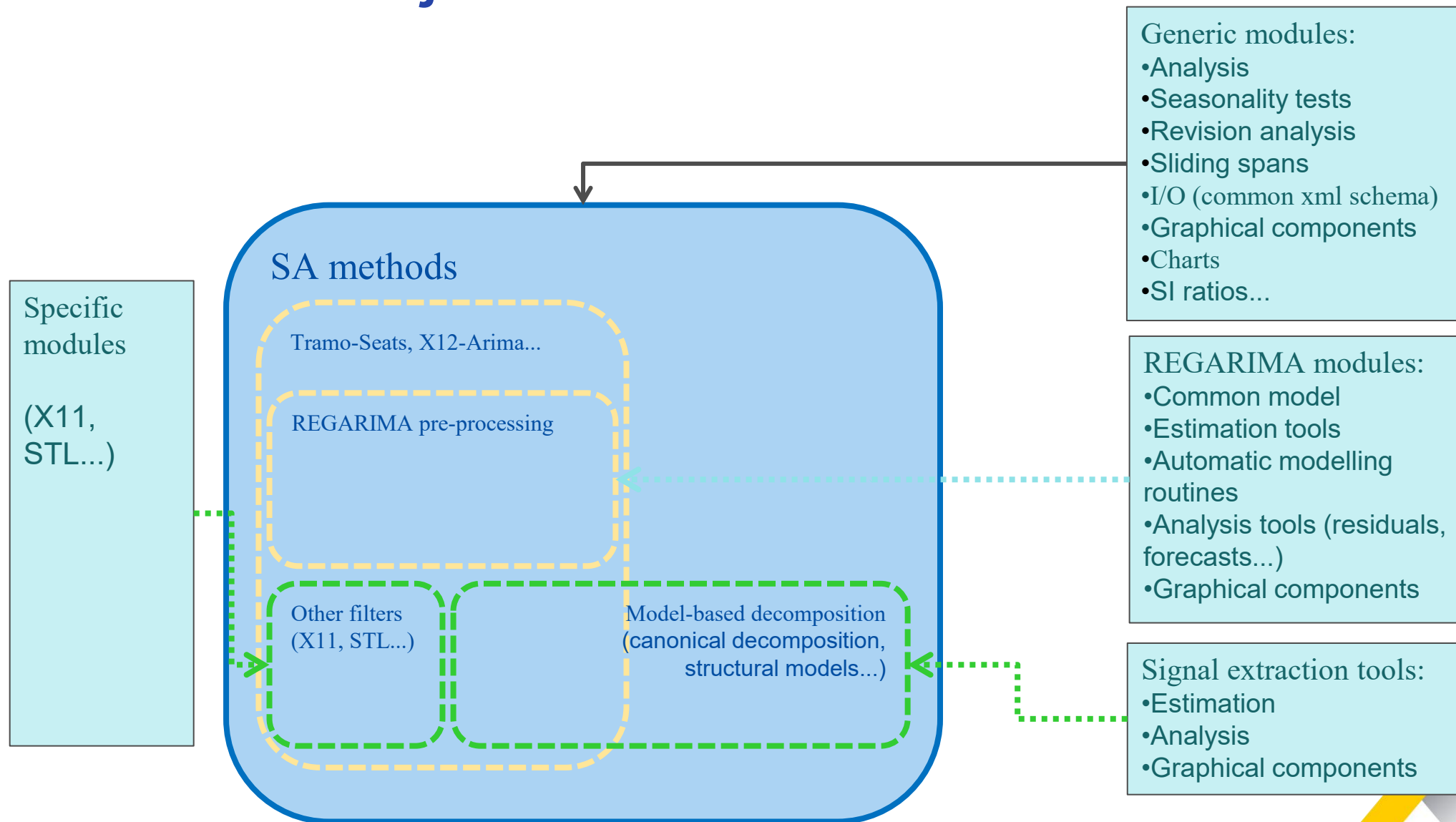
- Originally based on:
- Tramo-Seats (BDE)
- X12-Arima (USCB)

<https://github.com/jdemetra>

3. Statistical content



4. Seasonal adjustment framework



5. Final remarks

- JD+ is a complete re-factoring of Tramo-Seats and of X12-Arima in an open OO framework. In some cases, the new algorithms may lead to (usually slightly) different results .
- JD+ is also designed for the handling of related time series problems, especially through a rich state space library.
- By developing it as an open-source solution, we have tried to create an environment appropriate to external collaborations.

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